PROBLEM SET 0

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 60)

1) (a) **(5 pts)** Verify Eq. (12.4.2) by first constructing the 3x3 matrices corresponding to $R(\epsilon_x \mathbf{i})$ and $R(\epsilon_y \mathbf{j})$, to order ϵ.

hint: start with writing the standard rotation matrices along the *x* and *y* axes in terms of the sines and cosines of rotation angle ϵ, and use the Taylor expansion to approximate them to first order, before verifying Eq. 12.4.2

(b) **(5 pts)** Provide the steps connecting Eqs. (12.4.3) and (12.4.4a)

hint: start with the general expression for the operators associated with an infinitesimal rotation, as expressed in Eq. 12.2.7; $U[R(\epsilon_z \mathbf{k})] = I - \frac{i\epsilon_z L_z}{\hbar}$

(c) **(5 pts)** Verify that L_x and L_y defined in Eq. (12.4.1) satisfy Eq. (12.4.4a).

(d) **(5 pts)** Verify that $[L^2, L_i] = 0$, for *i=x, y, or z*

2) (a) **(5 pts)** Verify that the 2x2 matrices $J_x^{(1/2)}$, $J_y^{(1/2)}$, and $J_z^{(1/2)}$ obey the commutation rule $[J_x^{(1/2)},\ J_y^{(1/2)}] = i\hbar J_z^{(1/2)}$

hint: the superscript represents to total angular momentum value j.

(b) **(5 pts)** Do the same as in part (a) but for the 3x3 matrices $J_i^{(1)}$

(c) **(5 pts)** Construct the the 4x4 matrices and verify that $[J_x^{(3/2)},\ J_y^{(3/2)}] = i\hbar J_z^{(3/2)}$

3) A particle is described by the wave function

$$\psi_E(r, \theta, \phi) = A e^{-r/a_0}, \quad (a_0 = \text{const.})$$

(a) **(5 pts)** What is the angular momentum content (l, m) of the state ?

(b) **(5 pts)** Assuming ψ_E is an eigenstate in a potential that vanishes as $r \to \infty$, find E.

(c) **(5 pts)** Having found E, consider finite r and find $V(r)$.

4) **(5 pts)** Provide the steps connecting Eq. (12.6.3) and Eq. (12.6.5)

5) **(5 pts)** Find the energy levels of a particle in a spherical box of radius r_0 in the $l = 0$ sector.